

ME 209 - Manufacturing Processes

Instructor: N Ramakrishnan & Rakesh Singhai

Course Info

Course credits: L T P C

2 0 1.5 3

Class Timings: 11.30AM - 12:50 PM; every Tuesday and Friday (Slot G1 and G2)

Classroom: AB7/209

Prerequisites

There are no prerequisites except strong fundamentals of general engineering principles and mathematics. Some basic coding or programming skills are also desired.

Learning Objectives

1. At the end of this course, you will understand all the manufacturing processes and also the suitability of each process for an objective. Process to be used depends on part material, its shape and features, surface finish and accuracy.
2. You will also learn about processes used for developing prototypes quickly as well as about so called Non-traditional manufacturing processes ,which are expensively used in making sophisticated tools and also in electronic hardware manufacture.
3. You will learn about the economic aspects of manufacturing. Cost of manufacturing is important and depending upon the volumes required, processes to be used can be significantly different.

Teaching Methods

The course is primarily a lecture-based course along with laboratory work, assignments, quizzes, exams Active class participation during classes is both encouraged and expected. Class lectures will encompass real-life applications of learnings from the course. Some lectures will include problem-solving sessions.

You are encouraged to ask questions right in the middle of a sentence if you do not understand. Focus is on learning outcome. The course's overall aim is to expose you to all manufacturing processes. Assembly is also a manufacturing process.

There will be frequent quizzes to ensure that learning is happening. Regular attendance is extremely important.

Grading

Attendance	5%
Laboratory	25%
Assignments	10%
Quizzes	20%
Mid Sem Exam	20%
End sem Exam	20%

Honor Code

All students are responsible for knowing and adhering to the Honor Code of IIT Gandhinagar, as well as knowing the consequences of violating the honor code. Violations of the Honor Code may include cheating, plagiarism, dishonesty, forgery, lying, and threatening behavior. Any incidents of Honor Code violation may invite sanctions such as expulsion, suspension, and monetary fine and can result in an unceremonious declaration of a 'Fail' grade in the course. The Honor Code can be found at <https://iitgn.ac.in/students/honourcode>

Students with Disabilities

If you qualify for accommodations because of a disability, please let me know privately as soon as possible. All accommodation, as per the institute norms, will be provided.

Detailed weekly schedule:

Week	Topic	Instructor
1	Introduction to Manufacturing : History/Importance/Classification	NR

2	Introduction to Materials: Fundamentals/Behaviour/properties	NR
3	Heat treatment: Hardening/tempering, carburizing, annealing	NR
4	Casting Processes: Basics/Pattern/Core	NR
5	Forming Processes: Plastic deformation/ Hot Cold Working/Bulk Deformation/Sheet metal deformation	NR
6	Joining Processes: Welding/ Brazing/Soldering/Adhesive	NR
7 & 8	Machining Processes: Chip formation/Tool geometry/Turning,miling/ Drilling etc/ Surface finish/Economics of machining	NR/RS
9	Abrasive Machining	RS
10	Additive Manufacturing /Rapid Prototyping	RS
11	Polymer manufacture/ Extrusion/ Injection Moulding/ Compression Moulding/ Blow moulding	RS
12	Non-traditional manufacturing: EDM/ECM/Ultrasonic /Abrasive Jet	RS
13	Assembly operations: Bolting/Pinning/ Mass production techniques	RS

14	Read drawing/ selection of process/process planning/	RS
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